

Four indispensable planar balls violate the bound $n + 1$

Conjecture 00000009114 ziangni-sys

The source asserts that the Besicovitch multiplicity bound is always dimension plus one. We use the standard centered-ball selection formulation: from a given family of balls, one must choose a subfamily covering all their centers, with controlled overlap everywhere. Already a finite family in the Euclidean plane disproves the claimed bound.

Counterexample. *There is a family of four open balls in $\mathbb{R}^2$ such that every subfamily covering its centers has multiplicity four at the origin. Consequently the universal selection bound cannot be $2 + 1 = 3$.*

Proof. Let

$$c_0 = (1, 0), \quad c_1 = (0, 1), \quad c_2 = (-1, 0), \quad c_3 = (0, -1), \quad B_i = B(c_i, 11/10).$$

Each B_i is an actual open ball for the Euclidean norm. Every center has distance one from the origin, so

$$0 \in B_i \quad \text{for all } i \in \{0, 1, 2, 3\}.$$

For distinct centers, the squared distance is either 2 or 4. In particular it is at least $2 > (11/10)^2$. Hence no ball contains another ball's center, whereas every ball contains its own center:

$$c_i \in B_j \iff i = j.$$

Now let $I \subseteq \{0, 1, 2, 3\}$ index any subfamily covering all four centers. To cover c_i , the equivalence above forces $i \in I$. Thus $I = \{0, 1, 2, 3\}$. Its multiplicity function

$$m_I(z) = \#\{i \in I : z \in B_i\}$$

therefore satisfies $m_I(0) = 4 > 3 = \dim(\mathbb{R}^2) + 1$. The full family does cover the centers, so the obstruction is its unavoidable overlap, not absence of a covering subfamily. $\square$

Scope. This is a given-family selection counterexample: the centers form a bounded finite set and all radii are the same positive number. It also forms a Besicovitch family in the usual sense of a common intersection with no center in another ball. We do not replace the supplied balls by smaller balls. A different theorem about fine covers supplying arbitrarily small replacement balls has different quantifiers; our conclusion concerns the source's unqualified universal multiplicity assertion. No conclusion about the separate needle-shaped extremizer clause is needed.

Formal verification. The Lean project uses the actual space `EuclideanSpace ℝ (Fin 2)` and actual `Metric.ball` sets. It proves the dimension, exact center norms, separation, common point, full-cover existence, forced equality of every covering subfamily with the full family, and the genuine filtered-cardinality multiplicity. Its final theorem excludes every subfamily satisfying an everywhere dimension-plus-one bound. The project pins Lean 4.19.0 and Mathlib in its manifest; run `lake build` in `lean/`.